TOOLKIT

Colab Survival Guide

Everything you need to use Google Colab in this course.

Everything you need to use Google Colab in this course, on one page. Keep it handy for the full six weeks.

1. What is Colab?

Google Colab is a free website for writing and running Python. **Your laptop only shows the screen. Google's computer does the work.** Python, pandas, matplotlib, scikit-learn and statsmodels are already installed, so there is nothing to set up.

🧠 Think of it as a cyber café computer that lives inside Chrome: free, always ready, and it never runs out of fuel.

2. Getting in

Step	Do this
1	Use **Google Chrome** (other browsers work, Chrome works best)
2	Sign in with a **Google account**. If your work account blocks Colab, use a personal Gmail
3	Click the **Open in Colab** button on the day's exercise
4	**File → Save a copy in Drive.** A new tab opens. Work only in that tab

Your copies live in **Google Drive → My Drive → Colab Notebooks**.

3. The screen

Part	Where	What it does
▶ **Run button**	left of each code cell	runs that one cell
Code cell	grey box	Python goes here
Text cell	plain text	instructions and notes. Double-click to edit
📁 **Files panel**	left sidebar	files you downloaded or uploaded
{x} Variables	left sidebar	shows everything Colab currently remembers
≡ **Table of contents**	left sidebar	jump between parts of the notebook
Runtime menu	top menu	restart, run all, run before

4. Running cells

You want to	Do this
Run a cell and move to the next	**Shift + Enter**
Run a cell and stay on it	**Ctrl + Enter** (Mac: **Cmd + Enter**)
Run everything from the top	**Runtime → Run all**
Run everything above this cell	**Runtime → Run before**
Stop a cell that runs forever	click the ■ button, or **Runtime → Interrupt execution**
Start fresh	**Runtime → Restart session** (then run STEP 0 again!)

Golden rule: run cells **top to bottom**. A cell can only use what the cells above it created.

5. Colab's short memory

- Colab **remembers** every variable you create, until the session restarts.
- Sessions end if you are **idle for a while** (roughly 90 minutes) or after many hours of use. You'll see *"Runtime disconnected"*.
- After a restart or disconnect: **run STEP 0, then the cells above where you were.**
- Files in the 📁 Files panel are **deleted when the session ends**. Your notebook (in Drive) is safe.

6. Files and your own data

You want to	Do this
Use the course data	Nothing: STEP 0 and `data_path("name")` download it for you
Upload a small CSV	Drag it into the 📁 Files panel, then `pd.read_csv("yourfile.csv")`
Keep files between sessions	Mount Drive: `from google.colab import drive; drive.mount("/content/drive")`
Download your notebook	**File → Download → Download .ipynb**
Share with the trainer	**Share** button (top right) → add their email as *Commenter*

⛔ **Real NIGCOMSAT data does not go into Colab** unless your unit head has agreed in writing. Colab runs on Google's servers, outside NIGCOMSAT. The course data is synthetic, so it's fine.

7. The AI button (Gemini)

✅ OK	⛔ Never
"Explain this error to me"	Pasting real telemetry, RF records or network logs
"What does `.asfreq()` do?"	Customer, staff or contract details
Pasting course (synthetic) data	Anything your unit has not cleared

Not sure? Describe the *shape* of your data ("a table of hourly voltages with a timestamp column") instead of pasting it. And always **check what the AI tells you**: it can be confidently wrong.

8. Keyboard shortcuts worth learning

Shortcut (Mac: Cmd instead of Ctrl)	Does
Shift + Enter	Run cell, go to next
Ctrl + Enter	Run cell, stay
Alt + Enter	Run cell, insert a new one below
Ctrl + M, then B	New code cell below
Ctrl + M, then D	Delete cell
Ctrl + M, then M / Y	Turn cell into text / code
Ctrl + /	Comment or uncomment a line
Ctrl + Z	Undo typing

9. Errors you'll meet (and the fix)

Error	Meaning	Fix
`NameError: name 'x' is not defined`	Colab doesn't remember `x`	Run STEP 0 and the cells above. Check for missing quotes
`KeyError: 'col'`	No column with that name	Copy the exact name from `df.head()`
`SyntaxError`	Typo: missing `)`, `"`, `,` or `:`	Read the line slowly; Colab underlines the spot
`IndentationError`	Spaces at the start of a line are wrong	Line it up with the lines around it
`FileNotFoundError`	File isn't where you said	Use `data_path("name")`, or check the 📁 panel

Error	Meaning	Fix
`ModuleNotFoundError`	A library isn't installed	`!pip install name` in a cell (rarely needed in this course)
Runtime disconnected	Idle too long	Reconnect, run STEP 0 and the cells above

10. Five habits of people who never suffer in Colab

1. **Save a copy first.** Every time.
2. **Top to bottom.** No jumping around.
3. **STEP 0 after every restart.**
4. **Read the error's last line first.** It usually tells you exactly what's wrong.
5. **Restart & run all** before you say “I'm done”. If it runs clean from the top, it's really done.